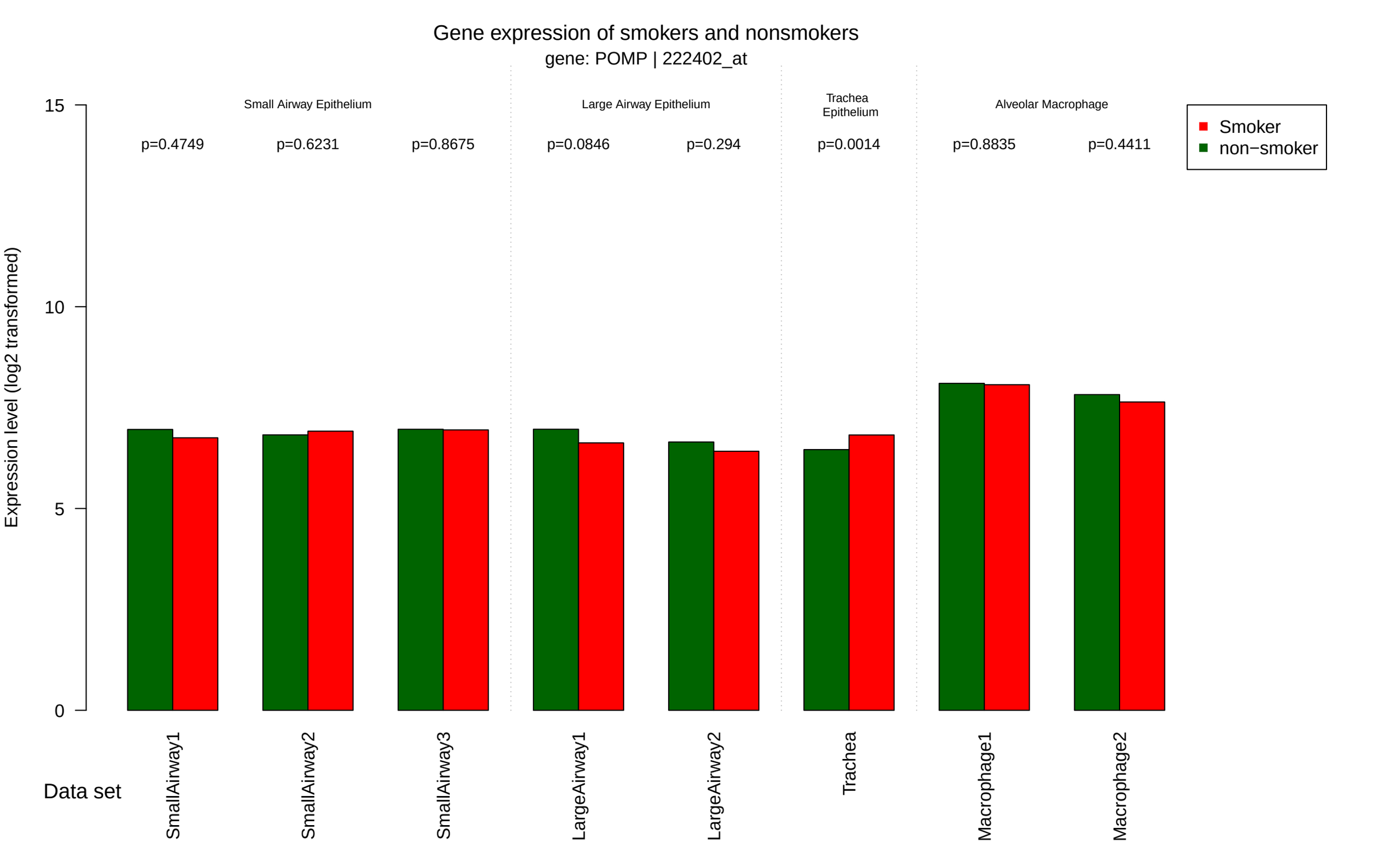
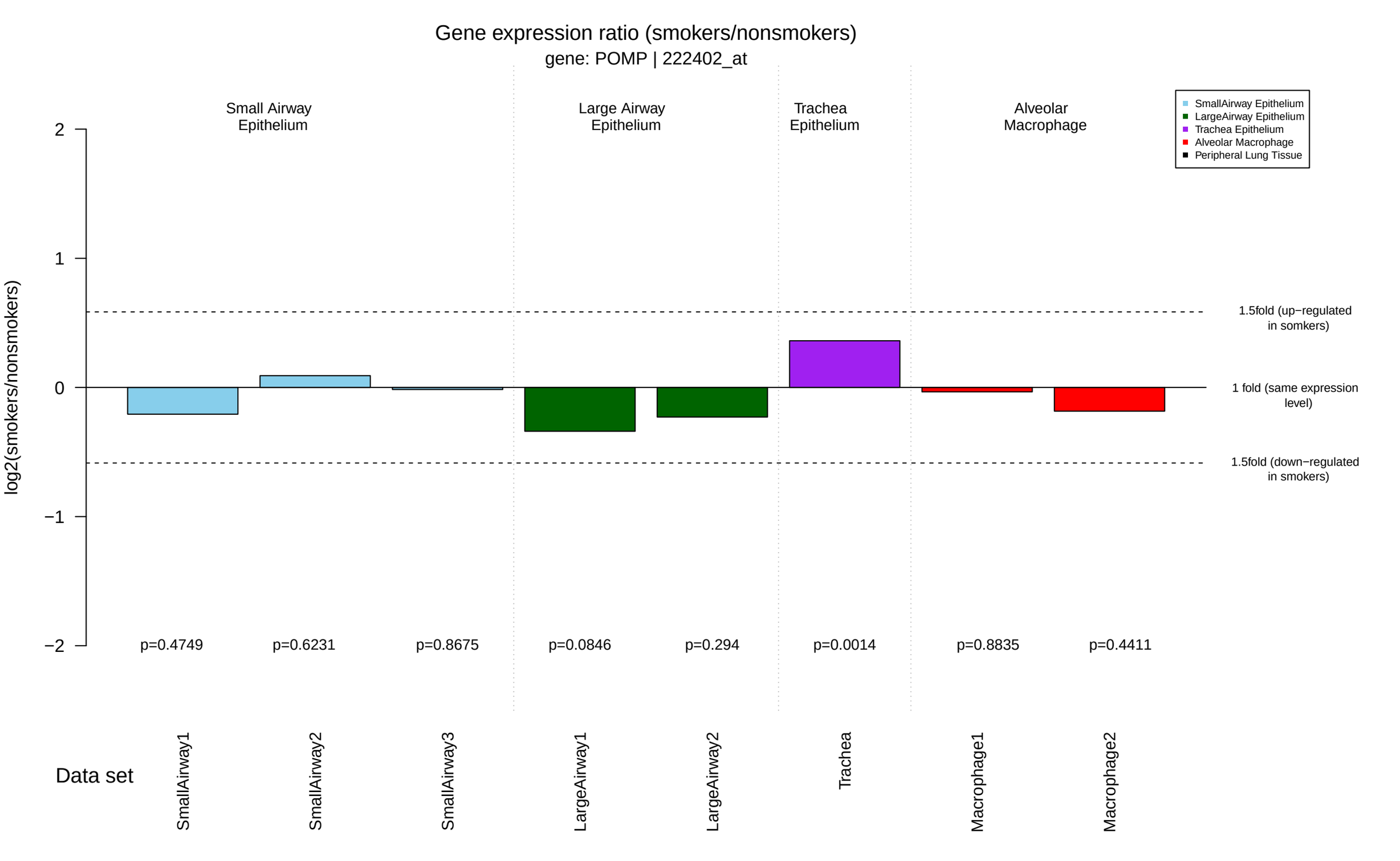
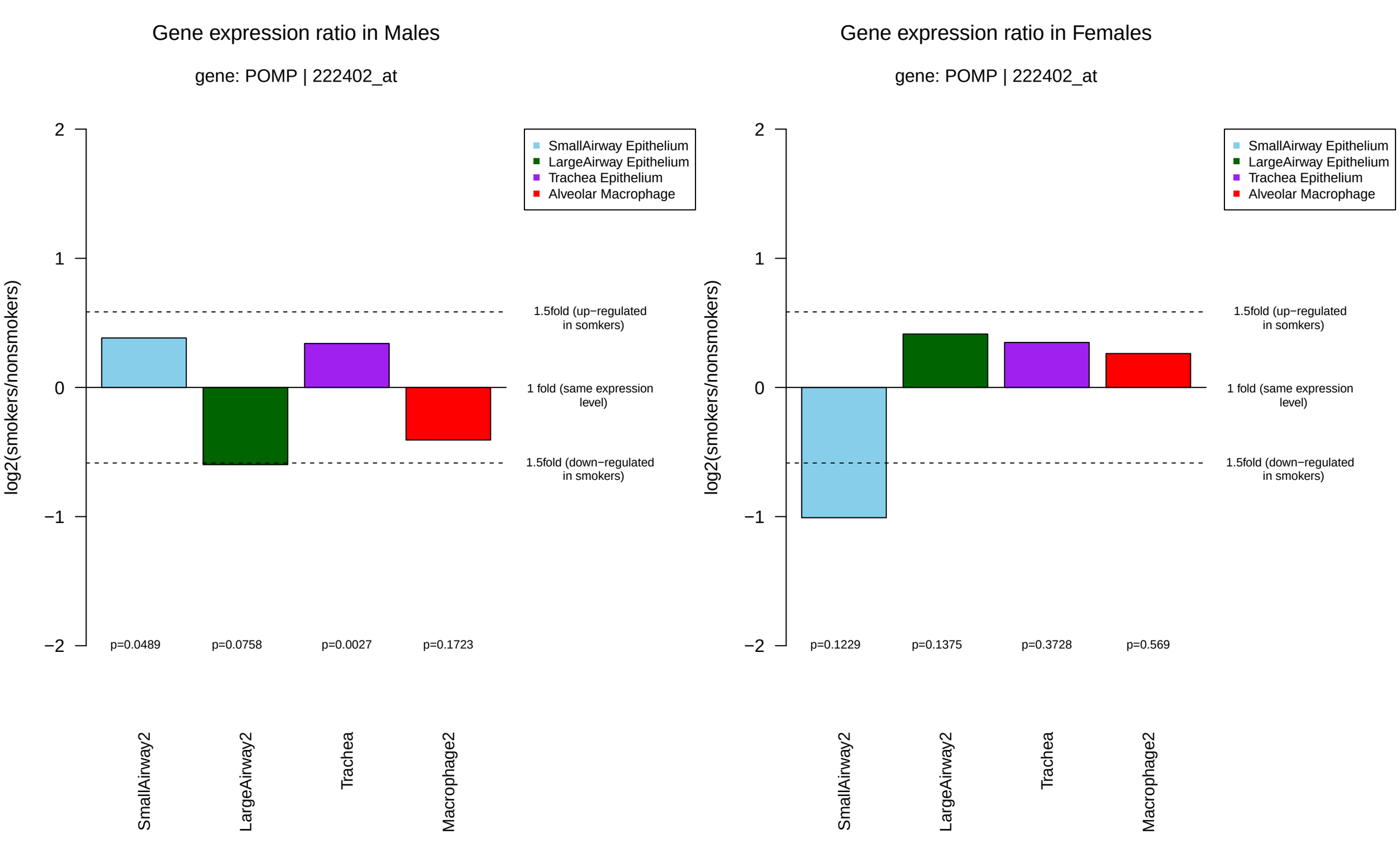
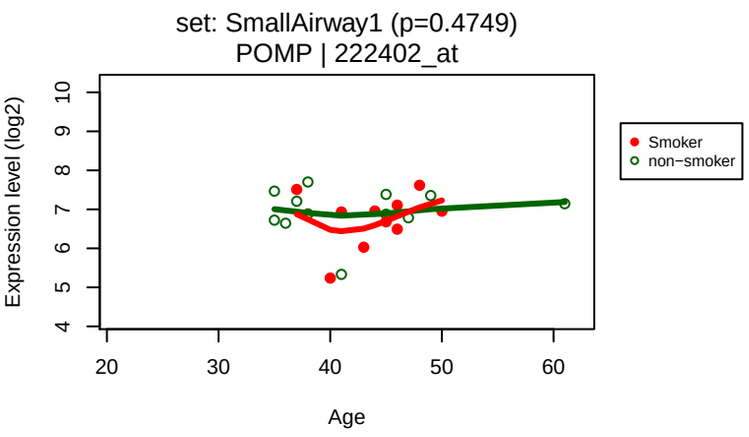
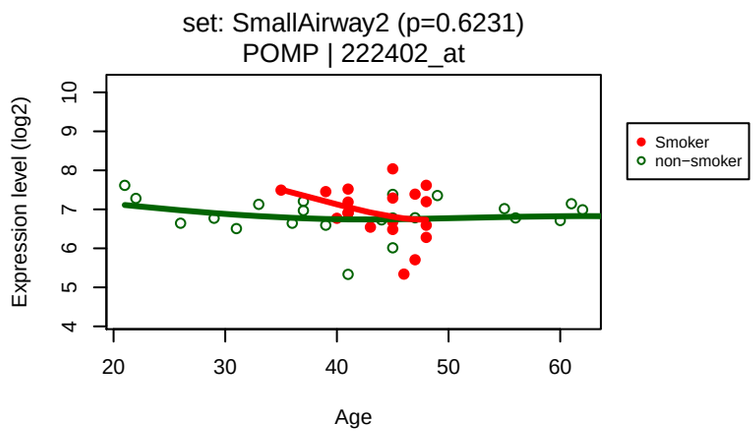
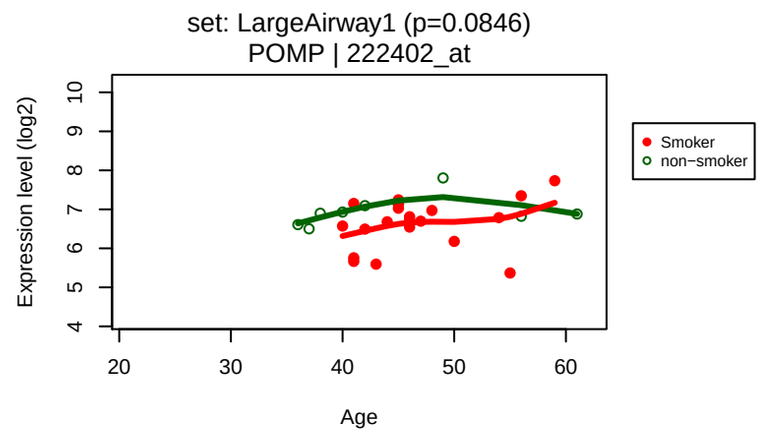

Gene expression of smokers and nonsmokers
gene: POMP | 222402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4749
p=0.6231
p=0.8675
p=0.0846
p=0.294
p=0.0014
p=0.8835
p=0.4411
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

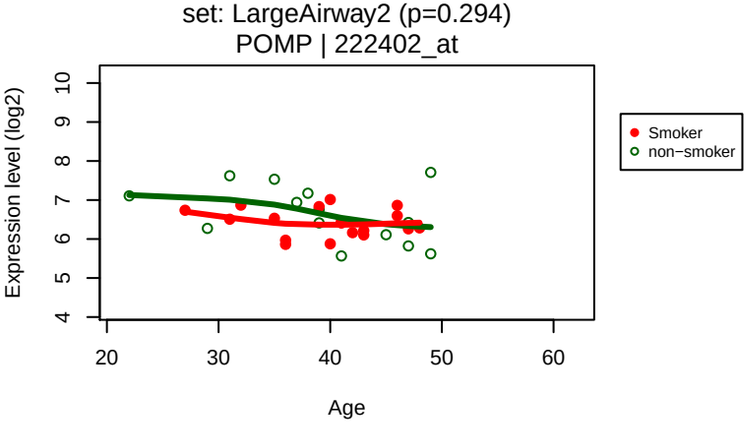

Gene expression ratio (smokers/nonsmokers)
gene: POMP | 222402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4749
p=0.6231
p=0.8675
p=0.0846
p=0.294
p=0.0014
p=0.8835
p=0.4411
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

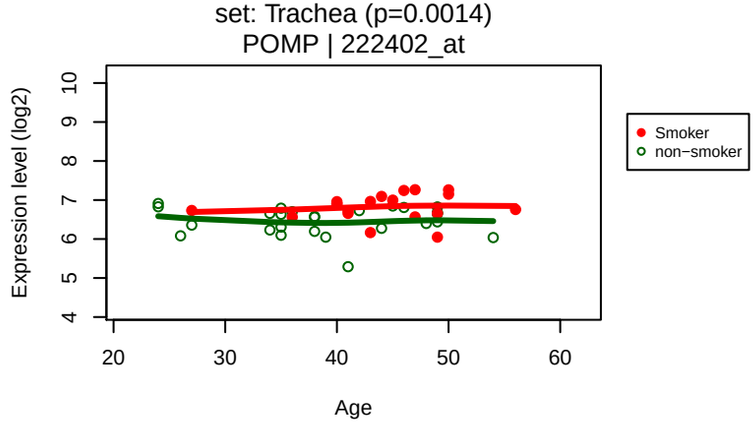

Gene expression ratio in Males
gene: POMP | 222402_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0489
p=0.0758
p=0.0027
p=0.1723
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POMP | 222402_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1229
p=0.1375
p=0.3728
p=0.569
SmallAirway2
LargeAirway2
Trachea
Macrophage2

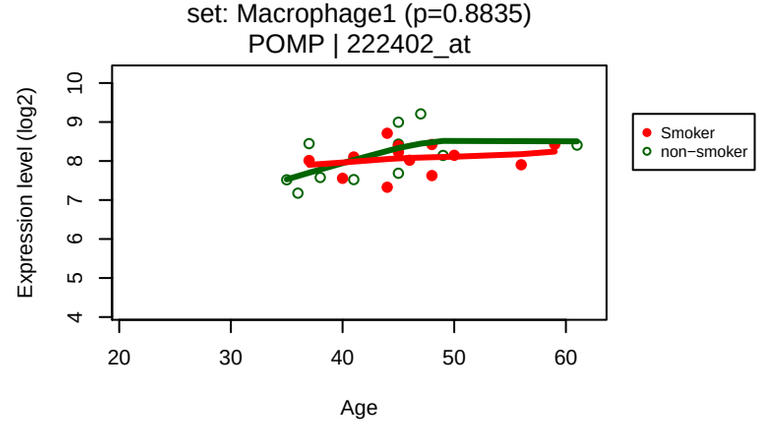

set: SmallAirway1 (p=0.4749)
POMP | 222402_at
Expression level (log2)
Age
Smoker
non−smoker

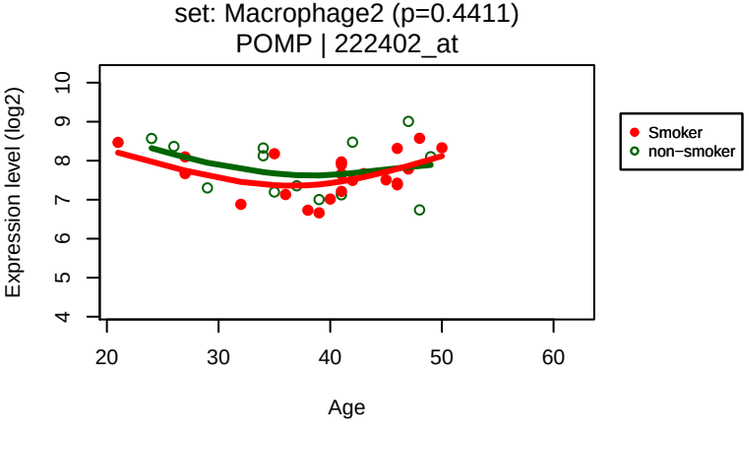

set: SmallAirway2 (p=0.6231)
POMP | 222402_at
Expression level (log2)
Age
Smoker
non−smoker

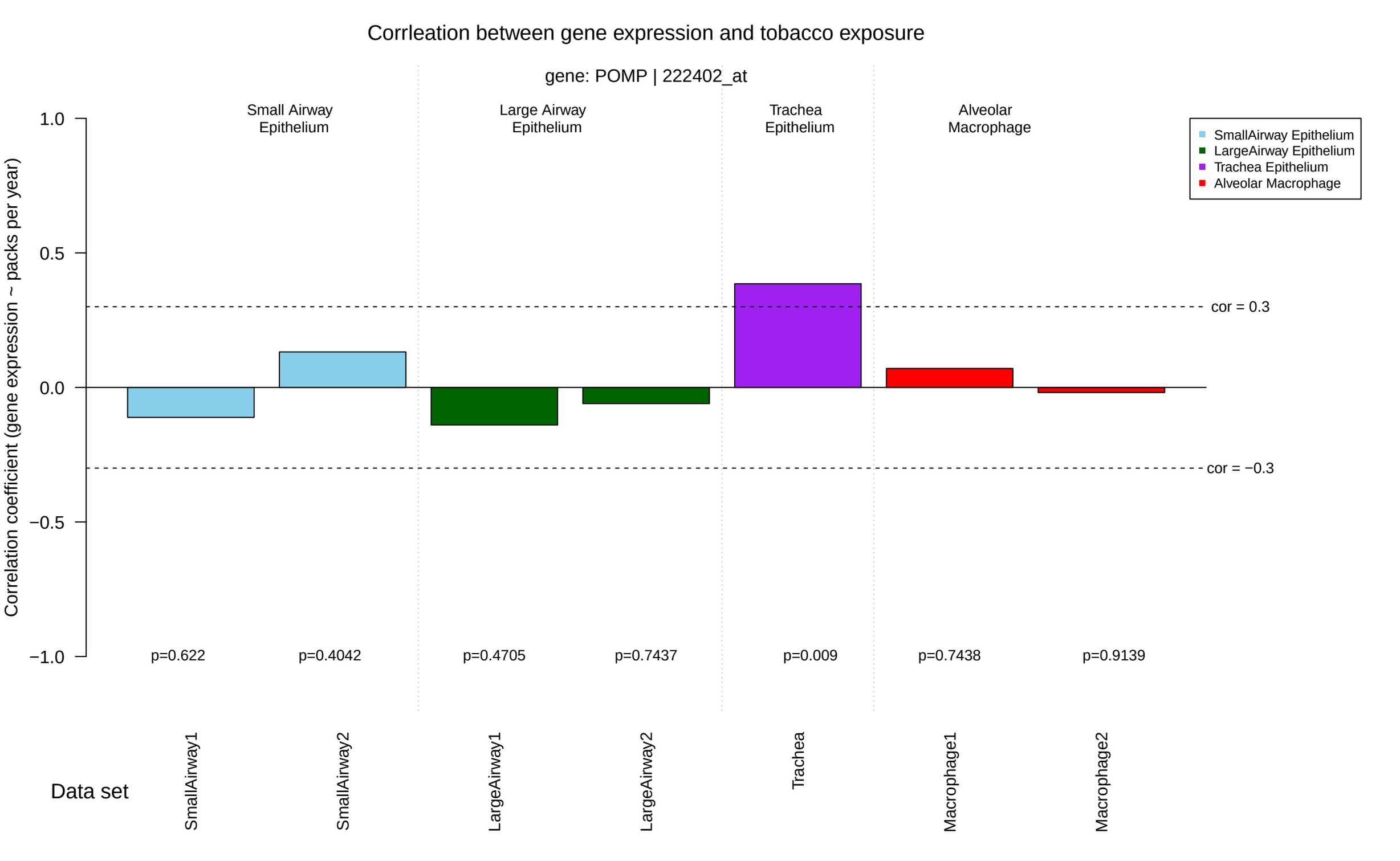

set: LargeAirway1 (p=0.0846)
POMP | 222402_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.294)
POMP | 222402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0014)
POMP | 222402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8835)
POMP | 222402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4411)
POMP | 222402_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POMP | 222402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.622
p=0.4042
p=0.4705
p=0.7437
p=0.009
p=0.7438
p=0.9139
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2